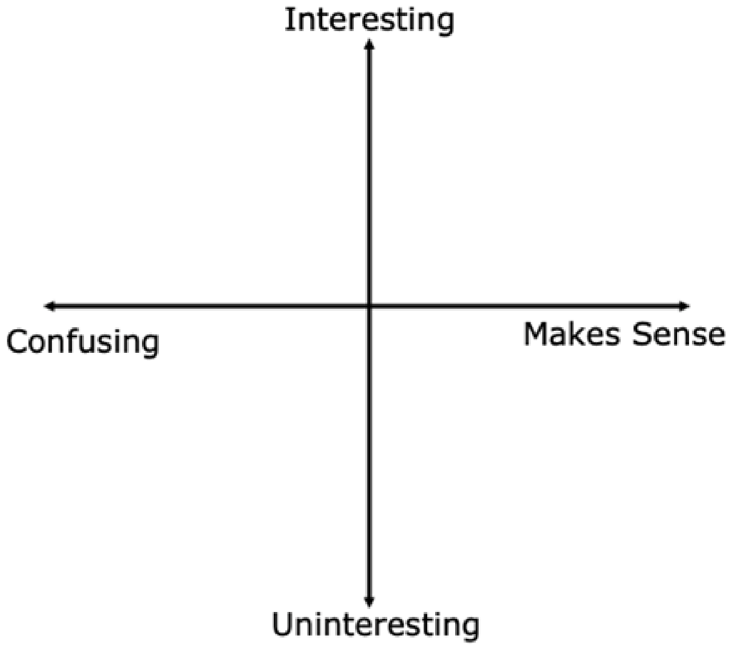
6.4.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and how much sense you made of today's learning target.



2. Explain why you chose that point and that quadrant.

Unit 6, Lesson 5: Graphing Two-Variable Linear Inequalities

Benchmark

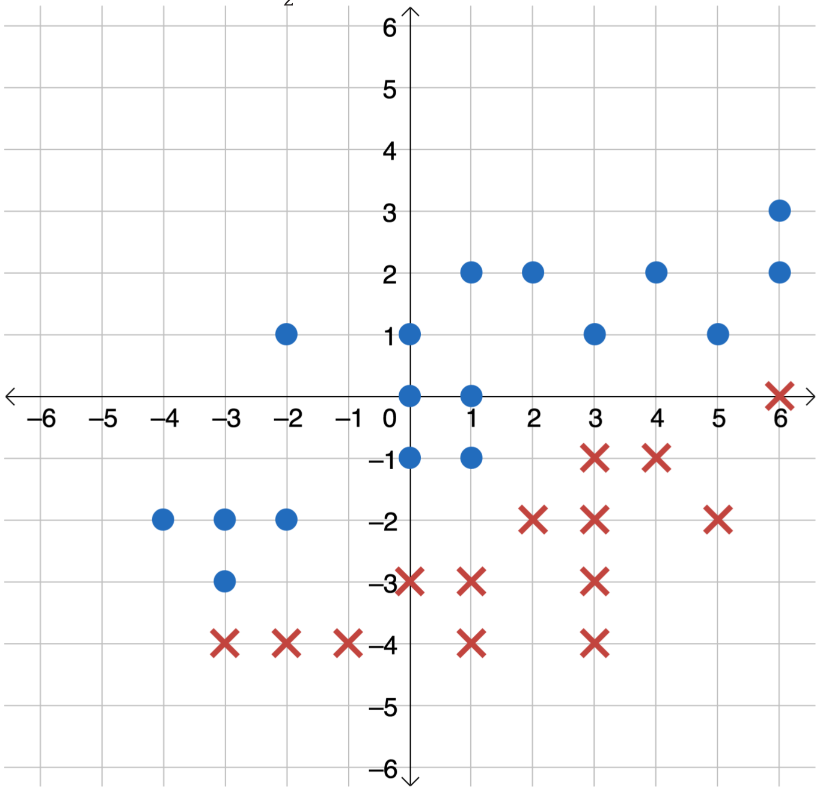
MA.912.AR.2.8 Given a mathematical or real-world context, graph the solution set to a two-variable linear inequality.

Learning Targets

- ☐ I can find the solution set of a linear inequality where one variable has a coefficient of 0.
- ☐ I can graph the solution set of a linear inequality given in standard form or point-slope form on a coordinate plane.

6.5.1 Warm-Up: Separating the X's and O's

1. Draw the line $y = \frac{1}{2}x - 2$ on the coordinate grid.



a. Which inequalities will include all of the circles?

$y \geq \frac{1}{2}x - 2$	$y \leq \frac{1}{2}x - 2$	$y > \frac{1}{2}x - 2$	$y < \frac{1}{2}x - 2$

b. Which inequalities will include all of the X's?

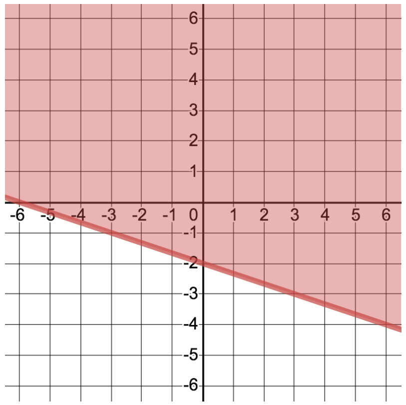
$y \geq \frac{1}{2}x - 2$	$y \leq \frac{1}{2}x - 2$	$y > \frac{1}{2}x - 2$	$y < \frac{1}{2}x - 2$

c. Draw an X on the graph so that only one of the inequalities will be true for all of the X's.

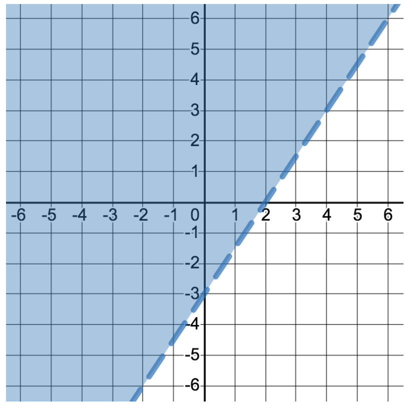
6.5.2 Collaboration: Two-Variable Linear Inequalities in Slope-Intercept Form

1. Four inequalities and their graphs are shown.

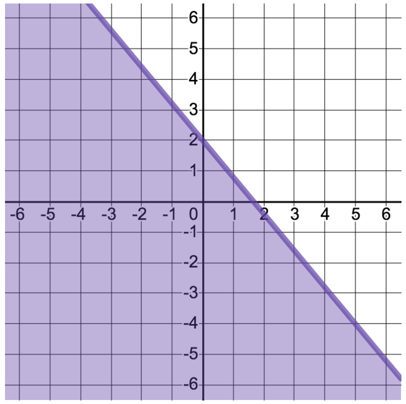
A $y \geq -\frac{1}{3}x - 2$



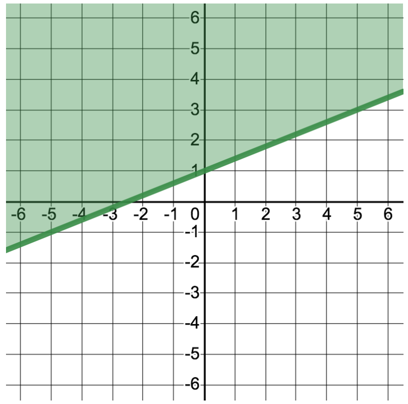
B $y > \frac{3}{2}x - 3$



C $y \leq -\frac{6}{5}x + 2$



D $y \geq \frac{2}{5}x + 1$



Complete the statements with the letter that corresponds to the inequality described.

a. Inequality _____ is the only inequality where the points on the boundary line are not included in the solution set of the inequality.

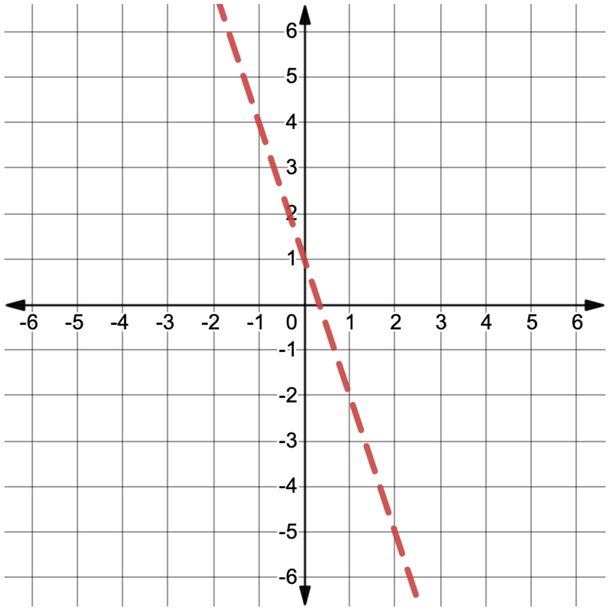
b. Inequality _____ is the only inequality where the points in the region below the boundary line are included in the solution set of the inequality.

2. Consider the inequality, $y > -3x + 1$.

a. Substitute $x = 2$ into the inequality $y > -3x + 1$ and evaluate.

b. What values of y satisfy the inequality when $x = 2$?

The boundary line, $y = -3x + 1$, is graphed on the coordinate grid. The ordered pair $(2, -5)$ is a point on the boundary line.



c. Discuss with your partner why the point $(2, -5)$ is not part of the solution set for the $y > -3x + 1$.

d. Write three ordered pairs with x-coordinates of 2 that are solutions to the inequality $y > -3x + 1$.

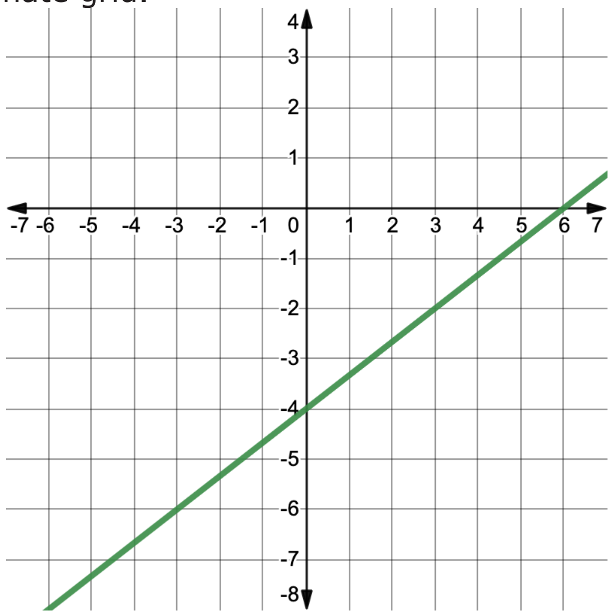
$(2,\_\_\_\_)$ $(2,\_\_\_\_)$ $(2,\_\_\_\_)$

e. Plot the three points you chose on the coordinate grid.

f. Shade the solution set of the inequality $y > -3x + 1$.

3. Consider the inequality, $y \leq \frac{2}{3}x - 4$.

The boundary line, $y = \frac{2}{3}x - 4$, is graphed on the coordinate grid.



a. Discuss with your partner why the boundary line is solid.

b. Select a value for x between -6 and 6.

c. Substitute your chosen value for x into the inequality $y \leq \frac{2}{3}x - 4$, and evaluate.

d. Based on your answer to Part C, shade the solution set of the inequality $y \leq \frac{2}{3}x - 4$.

e. Discuss with your partner how you can determine whether the graph of an inequality in slope-intercept form is shaded above or below the boundary line. Summarize your discussion.

6.5.3 Guided Instruction: Two-Variable Linear Inequalities in Standard and Point-Slope Forms

Linear inequalities in two variables can be written in forms other than slope-intercept form.

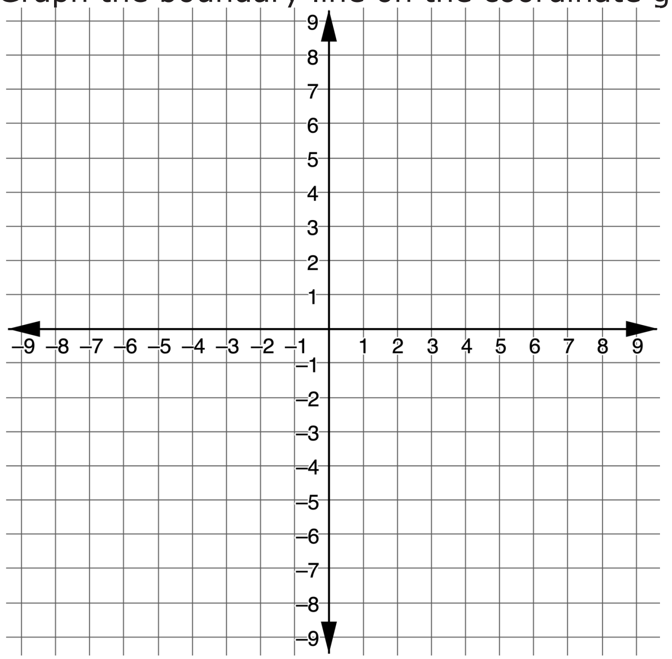
1. The linear inequality $2x - 3y > -9$ is written in standard form.

   a. Complete the statement.

   The boundary line of this inequality is ○ dotted ○ solid because the points on the boundary line are ○ included in ○ excluded from the solution set.

   b. Write the equation of the boundary line of this inequality in standard form.

   c. Graph the boundary line on the coordinate grid.



d. With your partner, discuss a conjecture about which side of the boundary line to shade to represent the solution set of the inequality.

e. Select two test points, one on each side of the boundary line. Substitute the ordered pairs into the inequality to determine which ordered pair is in the solution set.

Coordinates of Test Point		
Substitute the Coordinate Values Into the Inequality $2x - 3y > -9$		
Is it a solution to $2x - 3y > -9$**?**	○ Yes ○ No	○ Yes ○ No

f. Before testing any points, Elias made the following conjecture about which region would be shaded.

"I think I need to shade above the boundary line because the inequality sign is >, which means greater than."

Based on your work in Part E, explain whether Elias' conjecture was correct.

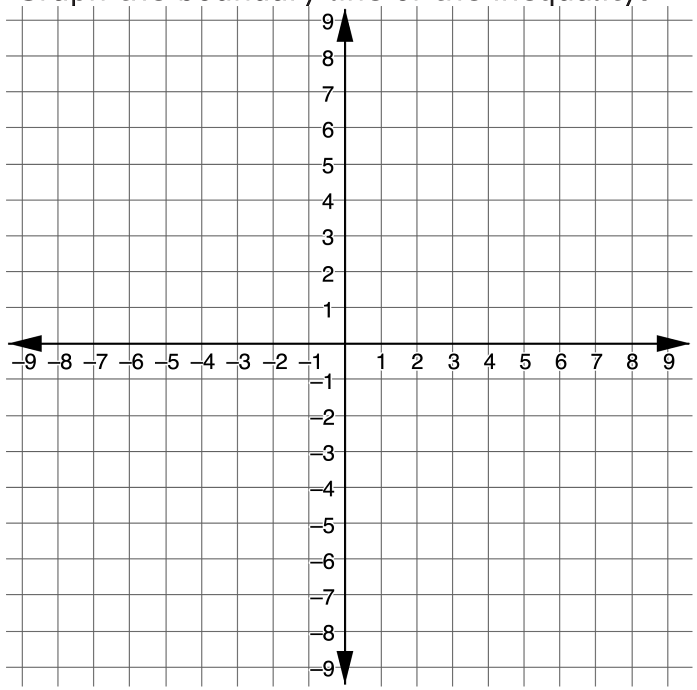
g. Shade the region on the graph that represents the solution set.

When graphing a linear inequality in two variables, **test points** can be selected and substituted into the inequality to determine which region should be shaded to represent the solution set.

2. Explain how it could be possible to determine which region to shade using only one test point.

3. The inequality $y - 8 \leq -\frac{1}{3}(x + 3)$ is given in point-slope form.

a. Graph the boundary line of the inequality.



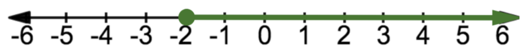
b. Select one test point that is not on the boundary line. Substitute your selected point into the inequality $y - 8 \leq -\frac{1}{3}(x+3)$ to determine whether it is in the solution set. Show your work.

c. Shade the region to represent the complete solution set of the inequality $y - 8 \leq -\frac{1}{3}(x+3)$.

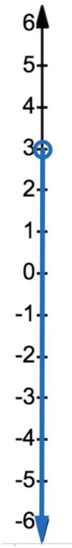
6.5.4 Collaboration: Inequalities With Vertical and Horizontal Boundary Lines

The graphs of the **one-variable** inequalities $x \geq -2$ and $y < 3$ are shown below and to the right.

$x \geq -2$



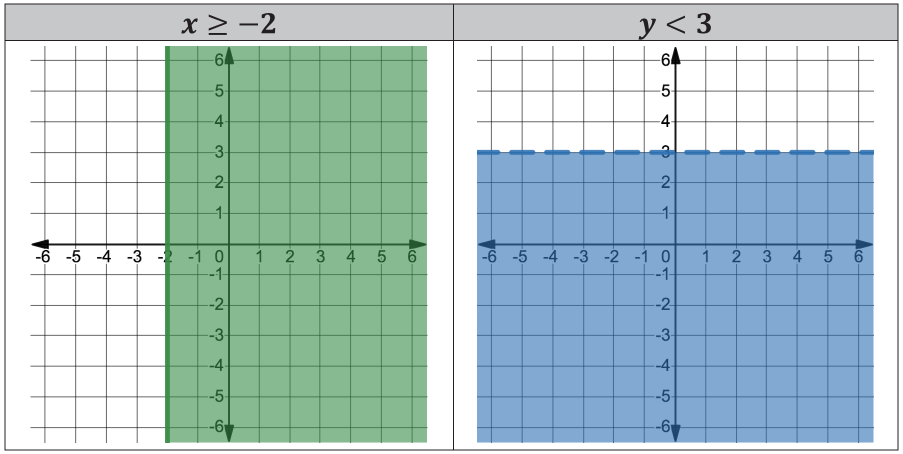
$y < 3$



The inequality $x \geq -2$ could also be a two-variable inequality, where the coefficient of y is 0.

Similarly, the inequality $y < 3$ could also be a two-variable inequality, where the coefficient of x is 0.

The inequalities $x \geq -2$ and $y < 3$ are shown graphed in **two-variables**.

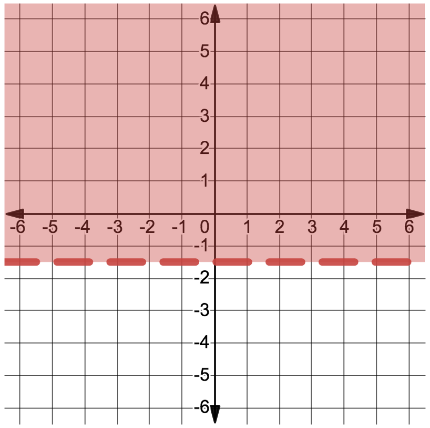
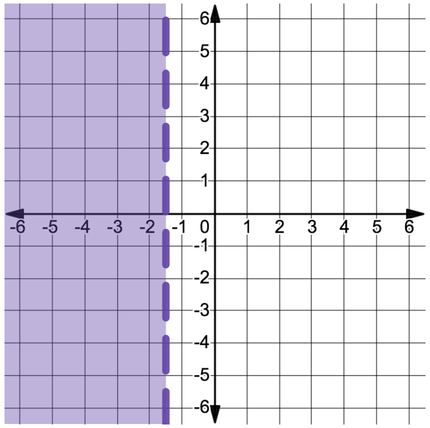
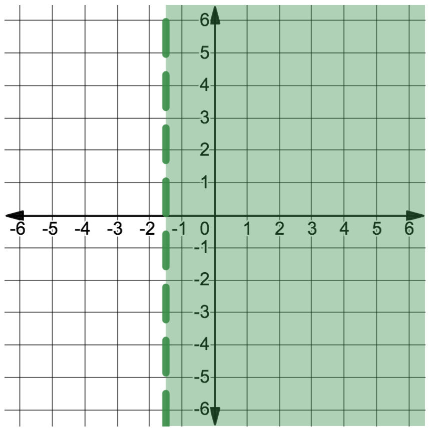
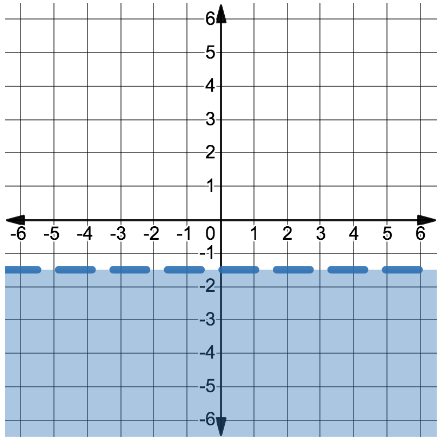


1. How do the two graphs in two variables relate to the corresponding graphs in one variable?

2. Four inequalities are given.

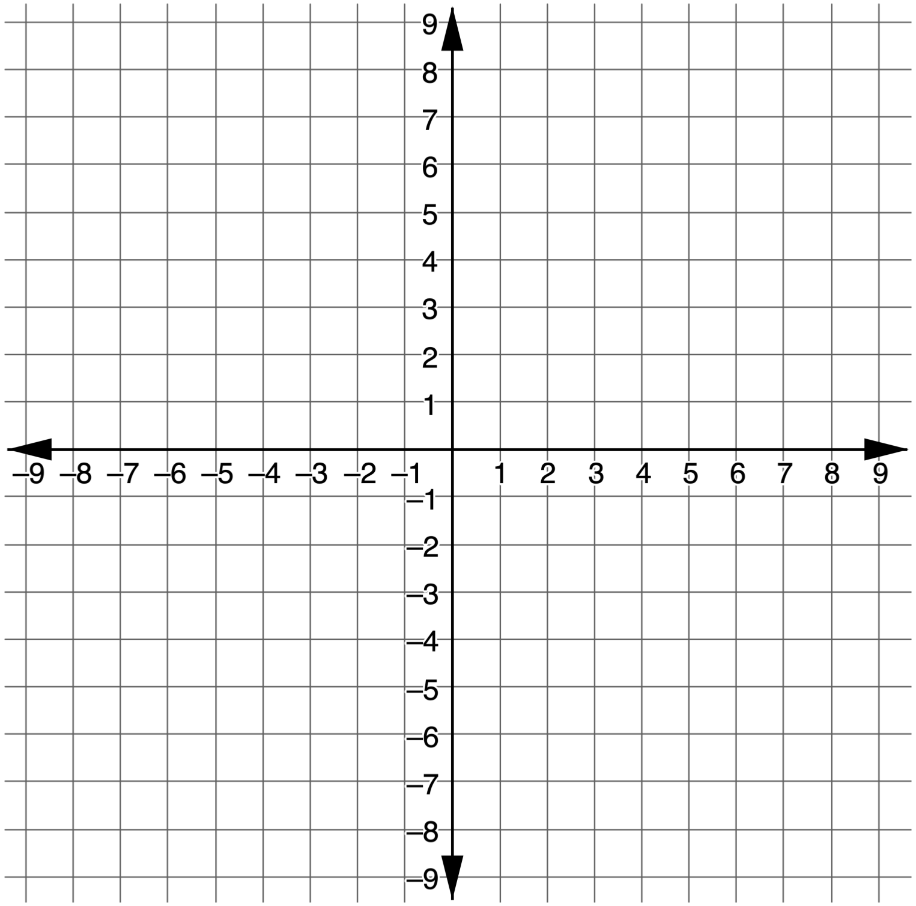
$x > -1.5$ $\quad$ $x < -1.5$ $\quad$ $y > -1.5$ $\quad$ $y < -1.5$

Match each inequality to its graph in two variables.

	
Inequality	**Inequality**
	
Inequality	**Inequality**

6.5.5 Wrap-Up: Ticket Out the Door

1. Graph the inequality $y + 5 > -2(x - 3)$.



2. Explain how you determined which part of the graph to shade.